MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

SECOND SEMESTER Skill Development Components (SDC) –

MG2SDCLMT101 - MATERIAL AND COLD CHAIN MANAGEMENT *(2025 ADMISSION ONWARDS)*

Maximum Marks: 50

Duration: 1 Hour 30 Minutes

Part A: Multiple Choice Questions

Answer all questions. Each question carries 1 mark. (marks)

1. The primary aim of material management is to:

- a) Increase production cost
- **b) Ensure uninterrupted production**
- c) Reduce employee strength
- d) Increase wastage

Answer: b) Ensure uninterrupted production

Explanation: Material management aims to provide the right quality and quantity of materials at the right time to prevent production downtime and bottlenecks.

2. Purchasing department ensures materials are purchased at:

- a) Highest price
- **b) Right price**
- c) Any price
- d) Selling price

Answer: b) Right price

Explanation: The "right price" is a core principle of procurement, balancing cost-efficiency with quality, reliability, and delivery speed.

3. Cold chain management ensures:

- a) Product spoilage
- **b) Quality preservation**
- c) Increased wastage
- d) High temperature

Answer: b) Quality preservation

Explanation: A cold chain is a temperature-controlled supply chain designed to preserve the safety, quality, and shelf life of perishable products.

4. Refrigerated containers are commonly known as:

- a) Dry vans
- **b) Reefers**
- c) Tankers
- d) Bulk carriers

Answer: b) Reefers

Explanation: "Reefer" is the standard shipping industry term for an intermodal container used in transportation that is refrigerated to move temperature-sensitive cargo.

5. In Human Resource Management, material management supports by:

- a) Reducing labor requirement
- **b) Providing safe storage facilities**
- c) Increasing absenteeism
- d) Eliminating training

Answer: b) Providing safe storage facilities

Explanation: Material management ensures that warehouses and storage areas are designed safely, thereby protecting the workforce handling those materials from workplace accidents.

6. A coding system must be:

- **a) Easy to understand**
- b) Difficult to interpret
- c) Secretive
- d) Unsystematic

Answer: a) Easy to understand

Explanation: To minimize errors in identification, picking, and stocking, a coding system must be logical, consistent, and easy for the warehouse staff to understand.

7. Banana carrier operations are mainly related to:

- a) Dry cargo transport
- **b) Temperature-controlled transport**
- c) Oil transport
- d) Coal transport

Answer: b) Temperature-controlled transport

Explanation: Bananas are highly climate-sensitive and must be transported in specialized, well-ventilated, temperature-controlled vessels or reefers to prevent premature ripening.

8. Air transport is preferred for perishable goods when:

- **a) Fast delivery is required**
- b) Delivery time is not important
- c) Cost is lowest
- d) Temperature is not needed

Answer: a) Fast delivery is required

Explanation: Air freight is used for highly perishable, high-value, or urgent goods (like medical supplies or fresh seafood) where minimizing transit time is critical.

9. XYZ analysis is based on:

- a) Unit price
- b) Annual consumption value
- **c) Value of inventory at a given time**
- d) Source of supply

Answer: c) Value of inventory at a given time

Explanation: While ABC analysis is based on annual consumption value, XYZ analysis classifies inventory based on the actual value of stock currently held in inventory (or its consumption stability and predictability over a timeframe).

10. Standardisation helps in:

- **a) Cost reduction**
- b) Increasing wastage
- c) Delaying production
- d) Increasing stock levels

Answer: a) Cost reduction

Explanation: Standardization reduces inventory variety, allows bulk purchasing, simplifies manufacturing, and reduces storage overheads, leading directly to cost savings.

Part B: Very Short Answer Questions

Answer any 10 questions. Each question carries 2 marks. (marks)

11. Mention the early methods of food preservation.

Early civilizations relied on natural preservation methods before mechanical refrigeration existed:

- **Sun Drying/Dehydration:** Removing moisture content from meats, fruits, and grains to prevent microbial growth.
- **Salting and Curing:** Drawing moisture out of food (especially meat and fish) via osmosis.
- **Fermentation:** Using beneficial microbes (like lactobacillus) to produce acids/alcohol that naturally preserve foods.
- **Smoking:** Exposing food to wood smoke, which contains compounds that act as natural preservatives and antimicrobial agents.
- **Root Cellaring / Ice Harvesting:** Storing food in cool underground spaces or using winter ice stored in straw.

12. Mention the types of refrigerated containers used in transport.

Refrigerated containers (reefers) are classified based on their temperature capabilities and design:

- **Standard Reefers:** Closed-type containers with an integrated, electric-powered cooling/heating unit.
- **Controlled Atmosphere (CA) Reefers:** Containers that regulate temperature and actively adjust oxygen () and carbon dioxide () levels to slow down the ripening of fresh produce.
- **Cryogenic/CO2 Temp-Controlled Reefers:** Containers that use dry ice or liquid nitrogen for cooling instead of mechanical compressors, often used for extreme-cold medical shipping.
- **Ventilated Reefers:** Designed for products like garlic or onions that do not need refrigeration but require continuous, fresh air circulation.

13. Mention the limitations of ABC Analysis.

ABC Analysis (Always Better Control) categorizes inventory into Class A (high value), B (medium value), and C (low value) based on annual consumption value. Its limitations include:

- **Ignores Criticality:** A cheap "Class C" item (like a specialized bolt) may be critical; if it runs out, the entire production line stops.
- **Based on Past Data:** It relies on historical consumption values, which may not accurately predict future spikes or changes in demand.
- **Monetary Bias:** It evaluates inventory solely on cost/value, overlooking factors like lead time, supplier reliability, or shelf life.
- **Requires Constant Updating:** Reclassifications are necessary as market prices and demand fluctuate, which requires ongoing administrative effort.

14. State the importance of shipment conformance in cold chain management.

Shipment conformance refers to ensuring that the goods arrive exactly in the agreed-upon state—including temperature, packaging integrity, and time windows.

- **Product Safety:** For biological drugs or food items, conformance prevents microbial growth or chemical degradation.
- **Regulatory Compliance:** Meets strict safety standards (e.g., FDA, HACCP) to prevent cargo rejection at borders or checkpoints.

- **Loss Prevention:** Prevents cargo damage claims, insurance disputes, and financial losses due to spoiled inventory.

15. State the importance of infrastructure in cold chain management.

Infrastructure forms the physical backbone of the cold chain.

- **Ensures Continuity:** Adequate infrastructure—including modern cold storage warehouses, pre-cooling facilities, reefer trucks, and power backups (generators)—prevents "breaks" in the cold chain.
- **Extends Product Shelf Life:** Specialized infrastructure preserves food and medicine quality from farms/factories to the end consumer.
- **Reduces Post-Harvest Losses:** High-quality local cold storage reduces food spoilage in agricultural supply chains.

16. Mention the function of refrigerated transportation in cold chain.

Refrigerated transportation is the dynamic link that moves products through the cold chain:

- **Temperature Maintenance:** Keeps products within their critical temperature range during transit (whether chilled or frozen).
- **Air Circulation:** Distributes cool air evenly throughout the cargo to eliminate hot spots.
- **Extended Reach:** Enables global and regional distribution of fresh produce and pharmaceuticals to distant markets.

17. State the importance of HML Analysis in inventory control.

HML (High, Medium, Low) analysis classifies inventory based on the **unit price** of the items:

- **Customized Purchasing:** Helps procurement departments apply strict buying controls over high-unit-price (H) items, while delegating low-unit-price (L) buying to lower staff tiers.
- **Security & Storage Control:** High-price items receive maximum physical security and strict inventory counts.
- **Cash Flow Optimization:** Prevents over-purchasing expensive units, keeping working capital fluid.

18. Explain the role of QR code and RFID coding in cold chain management.

Automatic identification technologies are crucial for tracking perishable shipments:

- **QR Codes:** Quick Response codes store extensive product data (such as batch numbers and expiration dates) that can be instantly scanned using handheld devices or smartphones to verify product authenticity at various checkpoints.
- **RFID (Radio Frequency Identification):** Active RFID tags can be embedded with temperature sensors. They transmit real-time temperature and location data wirelessly through container walls without requiring line-of-sight scanning, allowing logistics managers to identify thermal anomalies before cargo spoils.

19. Mention the role of pre-cooling in cold chain system.

Pre-cooling is the rapid removal of field heat from freshly harvested crops immediately after harvest, prior to transport or storage:

- **Slows Respiration:** Dramatically decreases the metabolic and respiration rate of fresh produce, delaying the ripening and aging process.
- **Limits Moisture Loss:** Prevents shriveling and weight loss by reducing transpiration.
- **Reduces Load on Reefers:** Reefers are designed to *maintain* temperature, not pull it down; pre-cooling ensures the cargo enters the transport vehicle already at the target temperature.

20. State the importance of proper layout in slaughterhouse operations.

A highly structured layout is mandatory in slaughterhouse operations due to biological hazards:

- **Prevents Cross-Contamination:** A clean linear layout separates "dirty" zones (stunning, bleeding, de-hairing) from "clean" zones (evisceration, chilling, packaging).
- **Optimizes Workflow:** Minimizes handling times, allowing carcasses to move swiftly into refrigerated units to prevent bacterial growth.
- **Ensures Humane and Safe Workspaces:** Provides safe channels for animal movement and ergonomic setups for workers handling sharp tools.

21. Define GOLF Analysis in Material Management.

GOLF Analysis is an inventory classification method based on the **source of supply** and procurement channels:

- **G (Government):** Items sourced from government undertakings, public sectors, or canalized agencies, which involve long lead times and rigid payment terms.
- **O (Ordinary):** Sourced from standard open-market suppliers with moderate lead times and negotiable terms.
- **L (Local):** Procured from local suppliers, offering immediate delivery and low transportation costs, requiring minimal safety stock.
- **F (Foreign):** Sourced from international suppliers, characterized by long lead times, customs clearance, currency fluctuations, and high shipping costs.

22. State the role of Purchasing Management in material control.

Purchasing management controls the inflow of external materials into the organization:

- **Cost Control:** Achieves savings through bulk discounts, negotiation, and favorable payment terms.
- **Inventory Optimization:** Works with warehouse teams to schedule deliveries (like Just-In-Time) to avoid overstocking or stockouts.
- **Supplier Relationship Management:** Cultivates a reliable supplier network to guarantee a steady, uninterrupted flow of high-quality raw materials.

Part C: Short Answer Questions

Answer any 4 questions. Each question carries 5 marks. (marks)

23. Describe the step-by-step method of classifying inventory items under ABC Analysis.

ABC Analysis divides inventory into three categories (A, B, and C) based on their annual usage value. The step-by-step classification process is as follows:

1. **Collect Inventory Data:** For every inventory item, compile the **unit cost** () and the **annual consumption quantity** ().
2. **Calculate Annual Usage Value:** Compute the total annual dollar value for each item using the formula:
3. **Sort the Items:** List all inventory items in descending order based on their calculated Annual Usage Value (highest value at the top, lowest at the bottom).
4. **Calculate Cumulative Values:** * Calculate the cumulative sum of the annual usage values.
    - Calculate the cumulative percentage of items (quantity-wise) and the cumulative percentage of usage value.
5. **Apply Classification Thresholds:** Group the sorted items into three categories based on Pareto's 80/20 rule:
    - **Class A:** Roughly **10% to 15%** of the physical items that account for **70% to 80%** of the total annual usage value. These require strict, close control, low safety stocks, and frequent reviews.
    - **Class B:** Roughly **20% to 30%** of the physical items that account for **15% to 20%** of the total annual usage value. These require moderate control and regular monitoring.
    - **Class C:** Roughly **50% to 60%** of the physical items that account for only **5% to 10%** of the total annual usage value. These require relaxed control, larger bulk orders, and high safety stocks.

24. Discuss how the coding system should be easily understandable and memorable by employees.

An effective material coding system must be designed for human usability, not just database efficiency. To make it easily understandable and memorable for employees, several key principles must be applied:

- **Logical Hierarchy (Mnemonic/Structured Coding):** The code should follow a logical progression (e.g., *Category -> Sub-category -> Specific Item*). For example, a code like ELEC-CBL-05 immediately tells an employee they are dealing with an **Elec**trical product, specifically a **C**a**bl**e, with a size or variant of **05**.
- **Use of Familiar Symbols & Mnemonics:** Avoid long, random strings of numbers (like 948105739). Instead, use letter abbreviations that sound like or represent the item (e.g., RM for Raw Materials, FP for Finished Products).

- **Consistency and Simplicity:** Keep code lengths uniform (e.g., always 6 or 8 characters). Mixing erratic lengths confuses warehouse staff and increases data entry errors.
- **Avoiding Ambiguous Characters:** Do not mix characters that look similar to the eye. For example, avoid placing the number zero (0) next to the letter O, or the number one (1) next to the letters I or L (1I0O).
- **Visual Aids and Training:** Standardized charts, color-coding corresponding to code blocks, and simple training manuals help employees memorize basic codification structures, reducing search times and picking errors on the warehouse floor.

25. Explain the managerial applications of FSN Analysis in material control.

FSN Analysis classifies inventory based on issue frequency and movement speed into **Fast-moving (F)**, **Slow-moving (S)**, and **Non-moving (N)** items. Managers apply this analysis to make critical operational decisions:

- **Warehouse Layout and Space Optimization:** Fast-moving (F) items are placed near loading docks, dispatch areas, or lower, easy-to-reach shelves to minimize travel time for picking staff. Slow-moving (S) and non-moving (N) items are relegated to back shelves or upper racks.
- **Obsolescence Management:** FSN analysis helps managers identify Non-moving (N) stock that has been sitting idle for long periods. Managers can then decide to write off, sell back, or liquidate this dead stock to free up valuable warehouse space.
- **Procurement Strategy:** Purchasing departments use FSN data to adjust order policies. Fast-moving items are ordered frequently in optimized batches (using EOQ), while slow-moving items are purchased only upon direct request or in minimal quantities.
- **Inventory Auditing & Cycle Counting:** Managers allocate audit resources efficiently. Fast-moving items, which are prone to inventory discrepancies due to frequent handling, are cycle-counted more frequently compared to non-moving items.

26. Describe how effective negotiation contributes to cost reduction and better supplier relations.

Negotiation is not a zero-sum game of trying to "beat" the supplier; rather, it is a strategic tool designed to balance cost savings with long-term partnership value.

Cost Reduction

- **Direct Price Discounts:** Through market research and volume leverage, buyers can negotiate lower per-unit prices.
- **Favorable Payment Terms:** Negotiating extended payment terms (e.g., Net 60 instead of Net 30) improves the buyer's cash flow and working capital.
- **Reduction in Total Cost of Ownership (TCO):** Effective negotiation targets "hidden" costs, securing concessions like free shipping, extended warranties, lower minimum order quantities (MOQs), and complimentary technical training.

Better Supplier Relations

- **Collaborative "Win-Win" Outcomes:** Good negotiation seeks outcomes where both parties benefit. When a supplier feels they have secured a fair margin, they remain committed to quality.
- **Open Communication Channels:** The process of negotiation establishes clear lines of communication, aligning expectations regarding quality, delivery schedules, and dispute resolutions.
- **Priority Status:** A professionally negotiated contract builds trust. In times of raw material shortages, suppliers will prioritize buyers with whom they share a strong, mutually respectful relationship.

27. Describe the role of retail outlets and point-of-sale storage in the cold chain.

Retail outlets and point-of-sale (POS) storage act as the "last mile" and final custody points of the cold chain before products reach the consumer.

- **Maintaining the Cold Chain Link:** Once refrigerated trucks deliver goods to a retail store, the products must be transferred immediately to POS refrigeration units (such as display freezers, open chillers, or back-room cold storage rooms) to prevent thermal bridging and product degradation.
- **First-In, First-Out (FIFO) Enforcement:** POS storage layouts are organized to support FIFO/FEFO (First-Expired, First-Out) inventory rotation, ensuring older stock is placed at the front of the shelf for consumers to buy first.
- **Visual Quality and Customer Trust:** Well-regulated display units preserve the appearance, texture, and nutritional value of fresh meats, dairy, and produce, ensuring consumer safety and satisfaction.
- **Temperature Monitoring & Alerts:** Modern POS display cases are equipped with alarms and digital sensors that alert store staff if a freezer door is left open or if the temperature rises above safe thresholds, preventing large-scale food spoilage.

28. Describe the significance of weight distribution in reefer container stacking.

Proper weight distribution is critical when loading and stacking cargo inside a refrigerated (reefer) container. Its significance includes:

- **Enabling Proper Airflow:** Reefers cool their cargo by blowing cold air up through a T-bar floor. If weight is distributed unevenly, or if cargo is stacked too tightly in one area, it blocks the flow of air. This creates "hot spots" where heat builds up, leading to localized spoilage of perishable goods.
- **Preventing Structural Overload:** Reefers are built with insulated walls and specialized floor systems. Concentrating heavy weight in a small area can crack the floor or warp the interior walls, ruining the container's thermal efficiency.

- **Ensuring Vehicle and Vessel Stability:** When reefers are loaded onto trailer chassis, trains, or container ships, an uneven center of gravity can cause safety hazards during transport. During sharp turns or heavy seas, a poorly balanced container can tip over, resulting in catastrophic losses.
- **Reducing Shifting and Physical Damage:** Distributing the weight evenly and using proper dunnage ensures that boxes do not slide, crush, or fall over during transit, preserving both the packaging and the delicate items inside